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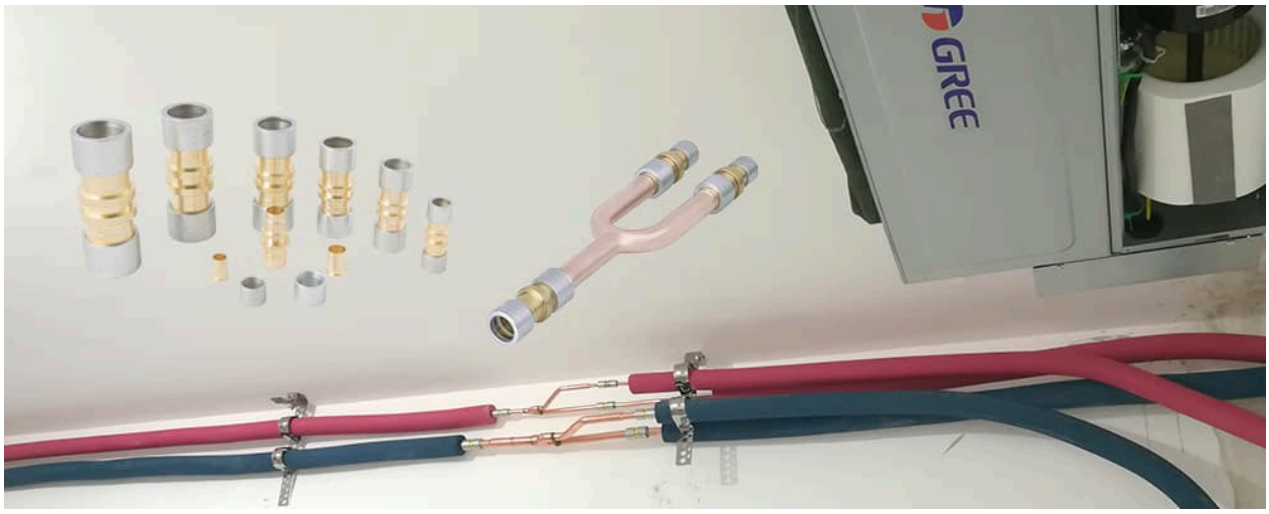
What is the VRF HVAC system- A Complete Guide?



agine you have an office building where the sun-facing rooms are warmer and need cooling, while shaded rooms on the other side need heating.

Privacy - Terms

Which air conditioner would you choose?

English ▾

We are introducing you to the VRF technology that solves all traditional chiller-based systems' drawbacks and ensures efficient and cost-effective benefits.

To give you a full understanding of the VRF system, we answer all your questions on mind about what the VRF system is in HVAC, what the VRF System definition is, what VRV is, what HP is in the VRF system, and everything you need to know.

In addition, we offer you the latest VRF installation technology(braze-free connection). Please view the HVAC press fittings.

What is Variable Refrigerant Flow?

VRF System stands for Variable Refrigerant Flow in HVAC, famous for its flexibility, efficiency, energy savings & cost-effective installation.

Let's explain the VRF system meaning in detail.

A VRF refrigerant flow system is the opposite technology of a traditional fixed-speed compressor, where an outdoor unit is connected to several indoor units, and the flow of refrigerant is adjusted dynamically to match the exact cooling or heating requirements of different zones in a building.

:an independently provide zonal control of the precise temperature of multiple individual indoor units.

English ▾

How can VRF do it? Because, unlike the traditional system, VRF uses a single compressor system to control multiple indoor units.

Daikin, a Japanese Air conditioner brand in 1982, first introduced VRF system technology.

Later, in the early 2000s, it was introduced in the U.S. From then till now, VRF systems have gained traction across various sectors worldwide because of energy efficiency, zoning capabilities, and flexibility and scalability.

Check VRF brands: VRF Outdoor Units

What are VRV and VRF Systems?

VRV System

VRF System

The terms VRV and VRF are sometimes used interchangeably, which confuses people about what a VRV system is and what the difference is between a VRV system and a VRF system.

The term VRV stands for Variable Refrigerant Volume, which refers to the same technology as VRF system technology.

As the Japanese AC manufacturing company Daikin invented this technology first, they trademarked this name.

Later on, all other companies that developed the same technology on their own or copied s technology referred to VRF or Variable Refrigerant Flow technology.

What are the Types of VRF Systems?

English ▾

There are four types of Variable refrigerant flow systems, they are:

1. Heat Pump VRF System (VRF HP) or 2-pipe system
2. Heat Recovery VRF System (VRF HR) or 3-pipe system
3. Water-cooling system
4. Gas operated system

Heat Pump VRF System or 2-pipe system: A heat pump system, also known as a 2-pipe VRF system, uses two refrigerant pipes to flow heating & cooling.

In heating mode, indoor units function as condensers, but in cooling mode, they work as evaporators.

A VRF heat pump can't provide simultaneous heating & cooling without using a branch controller, which is located in between the copper branch piping network and condenser.

The fewer pipes and connections benefit it for fewer leaks, with less maintenance, which improves its overall lifetimes.

It is ideal for open-plan office buildings, retail spaces, and apartment and residential buildings where the uniform temperature is sufficient and the same mode of heating or cooling is needed throughout.

English ▾

Heat Recovery VRF System or 3-pipe System: A heat recovery VRF system, also called a 3-pipe system, works on one cooling, one heating, and one return line that is connected to the condenser unit.

The name "Heat Recovery" refers to its operation, transferring heat from one part of the building to another rather than simply venting it outside.

It is the process of capturing heat from the area where cooling is needed and re-purposing this heat in another area where heating is needed.

The heating recovery VRF system is highly energy efficient, reduces waste, and saves operational costs.

It is used in large office complexes with varying occupant preferences, hotels where different rooms may need different heating or cooling preferences simultaneously, and in hospitals where patient rooms, operating theaters, and administrative offices need variable temperature control.

Water-cooling system: Another VRF system based on cooling methods is the water-cooled VRF system, which has a water circuit connected to a cooling tower, which helps in transferring heat more efficiently.

In this type of VRF refrigerant flow system, the refrigerant absorbs heat from the indoor un transfers it to the water circuit, and releases heat through a cooling tower or gains heat fro the boiler.

It has higher heat capacity than air and better performance, especially for larger buildings where an air-cooled system might have some struggle.

Other applications are large commercial skyscrapers, complexes, factories, industrial plants, etc.

Gas-operated system: Designed with gas-driven engines, this gas-operated VRF system can use natural gases or propane to power the system instead of electricity.

It is also known as a gas-driven VRF system, mostly used where electricity is scarce, and can also provide heat recovery functions.

this-driven engine drives the refrigerant circulation through gas-driven, it is suitable for applications that are more cost-effective than electricity.

It has lower operational costs in areas where gas is more accessible and energy efficient, and it reduces electricity grid loads.

English ▾

How Does a VRF HVAC System Work?

A non-inverter compressor only has two states that are on or off, and it runs at full capacity when it's turned on.

Unlike a non-inverter system, an inverter in the VRF system allows the compressor to speed up or down to adjust to the temperature requirements.

The VRF outdoor unit houses the compressor and circulates refrigerant to the VRF indoor unit to adjust the amount of refrigerant flow based on real-time demand.

It uses inverter technology to modulate compressor speed, which enhances the energy efficiency of the VRF system.

VRF indoor units are terminal units installed in different zones or rooms of a building in various forms, such as wall-mounted, ceiling cassette, ducted, or floor-standing units.

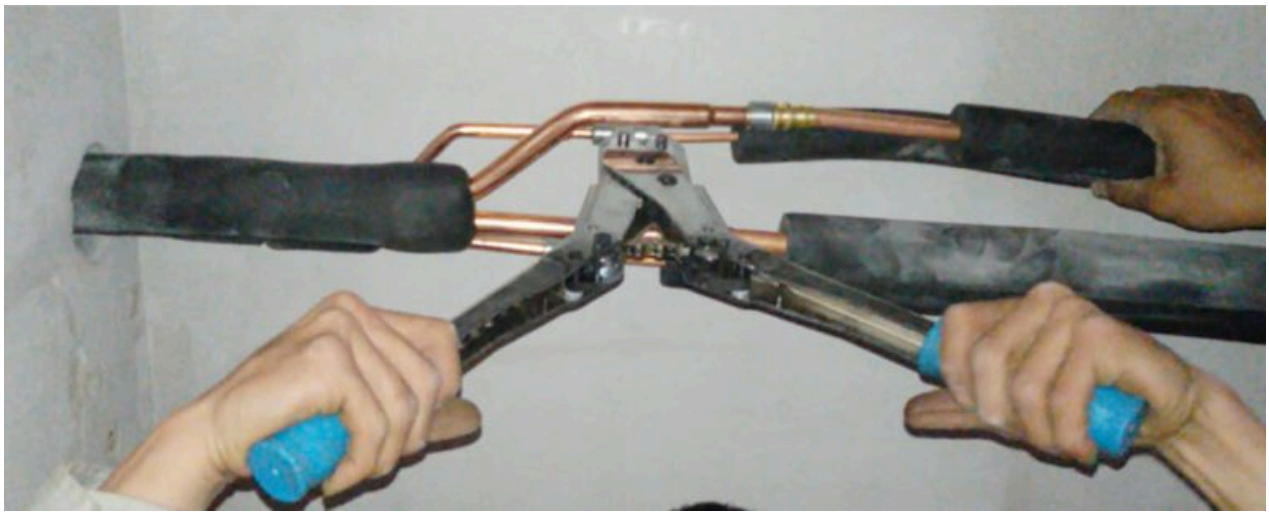
Each of these VRF units has its control system with individual temperature settings.

Based on the temperature control command, the refrigerant absorbs or releases heat as it passes through the indoor unit, providing the desired VRF heating and cooling.

Check the PDF file to know more.

English ▾

VRF System Advantages And Disadvantages:



Advantages of the VRF system:

VRF systems can use lokring tools and fittings, so the system pipes are cleaner and have ze leakage.

A variable refrigerant flow – VRF system has the most flexible design with expanded piping capabilities.

A VRF system doesn't have any ductwork or large pipe sizes, so it doesn't experience huge energy losses like another system that has ductwork.

It has diverse zoning capabilities with multiple types and sizes of VRF indoor units.

The more you can control the temperature, the more comfort you'll have.

The VRF system has the highest temperature customization settings.

High energy efficiency is another great advantage of a VRF system because its inverter technology doesn't cycle on and off but operates continuously at a part load.

Heat recovery VRF systems can also transfer heat from one room to another or vice versa, which saves energy and raises efficiency.

It is designed to run only as needed and under partial load conditions that make the VRF system less susceptible to any possible breakdown.

Let's compare VRF systems to traditional DX split systems or hydronic systems, like water source heat pumps, etc.

English ▾

VRF systems are much more saleable in expanding VRF heating and cooling system products.

Disadvantages of the VRF system:

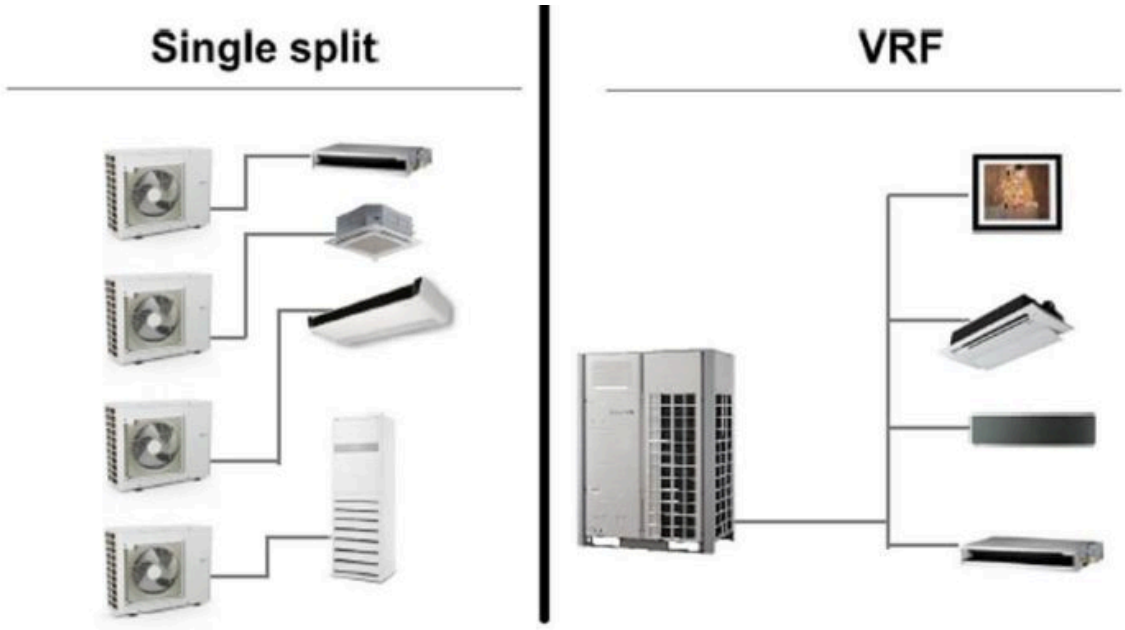
Having all of these advantages and benefits also brings one apparent disadvantage, which is the high primary cost.

In a heat recovery VRF system, piping is very complicated, and individual controllers in each VRF indoor unit cause more elements, parts, and setups that require higher installation & setup costs.

Its installation is complicated and time-consuming compared to other traditional split systems.

Variable refrigerant flow technology uses a larger volume of refrigerant than others, which also requires a high initial cost.

VRF System Vs. Split System:



When it's time to choose the right HVAC system for your building, understanding the difference between the VRF system and the Split system is important, as both systems have their benefits and drawbacks.

English ▾

Choosing either a VRF system or a Split system depends on your application's specific needs.

As we know, a VRF system is ahead of other counterparts in energy efficiency, flexibility, scalability, and comfort, with disadvantages like initial cost and complexity.

On the other hand, a split system consists of an outdoor unit and one or more indoor units with benefits such as cost-effectiveness, simplicity, and availability.

In a split system, each indoor unit operates independently, which leads to inefficiencies and more energy consumption as refrigerant flow needs to be adjusted based on real-time demand.

Based on the different features, functionality, pros and cons, VRF system and split system has different use applications.

Who should use a VRF system?

1. Large commercial buildings, large retail space
2. Hotels, motels, lodgings, office building
3. Apartments, dormitories, hospitals
4. Manufacturing plants, labs, restaurants
5. Mixed-use facilities
6. Multi-story buildings
7. Hospitality

Who Should Use Split Systems?

1. Small to medium-sized commercial business
2. Residential homes, small offices, retail spaces
3. A home without ductwork
4. Newer construction
5. Single-zone requirements
6. Small apartments

How to Install the VRF System?

English ▾

It's very crucial to follow the proper installation manuals of the VRF system manufacturer and maintain the quality of the installation.

Therefore reliable installation is the only way to effectively ensure the reliability of the VRF system.

Otherwise, many problems would arise. We have cooperated with Hisense Hitachi VRF to effectively solve the risk of leakage of the VRF system during installation.

Please check out our braze-free HVAC fittings to learn about the reliability of Miracle Ring.

Installing a VRF system involves many different parts and tests, from mounting the VRF indoor unit to doing refrigeration pipework, connecting Y branch copper fittings, etc.

After the installation of VRF indoor units and VRF outdoor units, refrigeration piping, and electrical wiring.

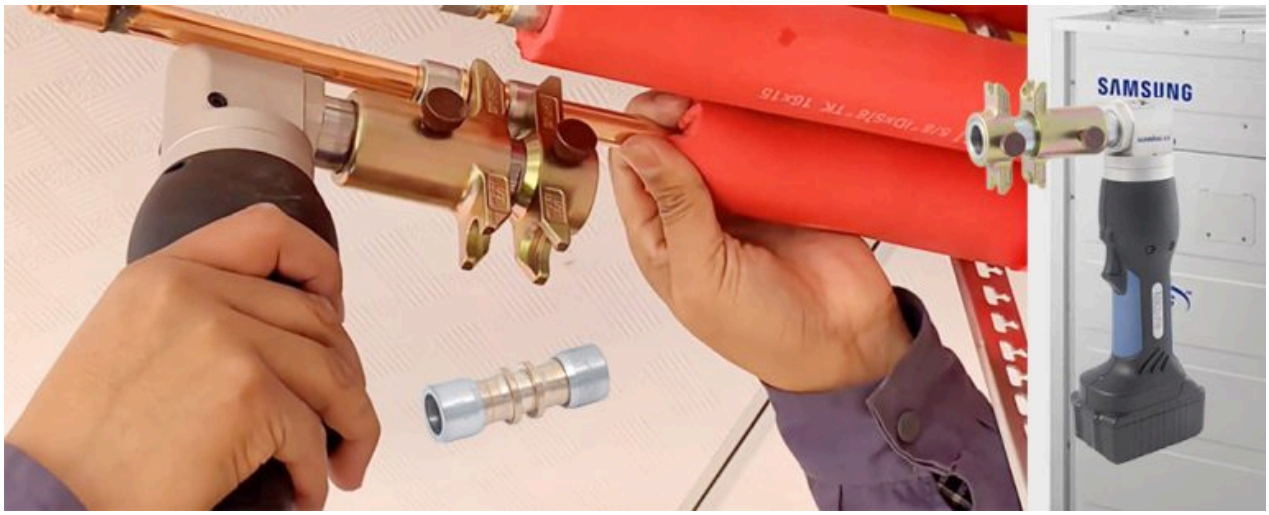
It's very important to run system configuration and testing like refrigerant leak test, pressure leak test, refrigerant charging, and verify electrical connections and power wiring.

When the final inspection is complete in the VRF system installation process, it's time to hand over the plant, provide maintenance training, and deliver relevant documentation, including system manuals, warranties, and maintenance schedules.

ad more about VRF system installation in our How to Install the VRF System guide to learn more.

English ▾

VRF HVAC System Cost:



Some say the VRF system installation cost is higher than the other alternative system, while others say it's super cost-effective; which one is correct?

If you install a VRF system, the cost will be approximately 5 to 20% higher than the equivalent capacity of other alternatives like DX system, water or air-cooled chilled system, etc, according to some studies.

The reason might be the long refrigerant piping, several indoor evaporator exchanges with their controlling system, etc.

In addition, the largest expense of installing a VRF system is the cost of personnel.

Therefore, using as few technicians as possible can be an effective way to save money. View our pipe press tools and fittings now.

Miracle flame-free joining technology is the best joining method for VRF systems.

Another study by GSA says that the installation cost of a VRF system can be the most cost-effective in a 10k to 80k square foot building, which can be $18 per square foot.

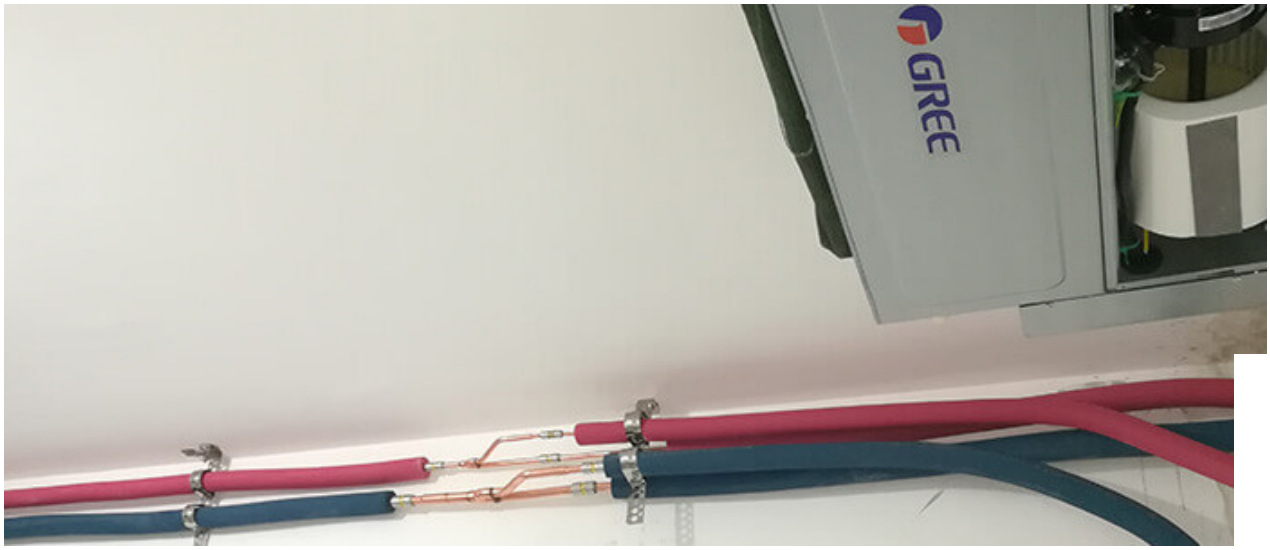
But in reality, it's tough to estimate, as the VRF system cost is dependent on the project and highly variable like the system size, capacity, whether it's a heat pump VRF system (HP) or a heat recovery VRF system (HR), number of VRF indoor units, installation complexity, the VRF system brand, and quality.

You also need to consider whether the application is small to medium commercial buildings, large commercial buildings, or residential applications.

English ▾

To learn more about the cost of the VRF system and installation cost, please read our article about **VRF system Costs**.

What Are The VRF System Manufacturers?



Before buying the VRF system for your project, finding the best VRF system brand & manufacturer is very important.

The market size was US $ 22.39 billion in 2022 and is expected to reach nearly US $ 52.04 billion.

Some world-class VRF technology manufacturers who are pioneers in VRF cutting-edge technology innovation are Gree, Media, Daikin, Haier, Hisense, AUX, etc.

Based on their reputation, top products, and innovation, there are some raking and list of VRF system manufacturers by a top VRF braze-free technology manufacturer.

You can browse this list based on the top VRF system manufacturers in different regions, like the Top 10 VRF manufacturers in China, the top 10 VRF system manufacturers in the USA, the Top-selling VRF manufacturers in Japan, and the top VRF manufacturers in Korea.